WEEK 6

Learn Online

Metal Welding

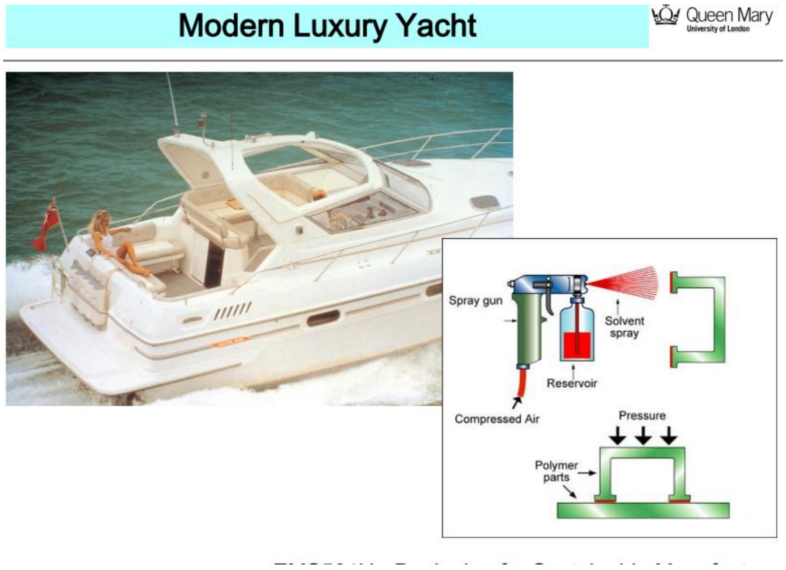
1. **Oxyacetylene Welding**
   - Uses a flame from an oxygen-acetylene gas mixture.
   - A filler rod is melted into a weld pool on the parent metal.
   - Common for manual, portable welding tasks.
2. **Metal Inert Gas (MIG) Welding**
   - Uses a consumable wire electrode with a gas shield (usually Argon).
   - An electric arc melts the electrode to form the weld.
   - Suitable for automated and high-speed welding.
3. **Electron Beam Welding**
   - A focused electron beam melts the metal in a vacuum.
   - Deep and precise welds, with minimal distortion.
   - Requires close fit-up; used in aerospace/nuclear sectors.

🧱 **Ceramic Welding**

4. **Diffusion Bonding**
   - Applies high pressure and temperature in a furnace.
   - Materials join without melting; atoms diffuse across the bond line.
   - Ideal for ceramics and dissimilar material joints.

🧵 **Polymer Welding**

5. **Induction Welding**
   - Uses an induction coil to generate heat in a metal insert.
   - Thermoplastics are fused around the heated insert.
   - Common for joining tubes in plastic systems.
6. **Laser Welding (Polymer)**
   - A laser beam focuses on a polymer sheet with a light-absorbing dye.
   - Heat melts and fuses the layers at the weld line.
   - Enables clean, precise polymer joints.



No External Bolts or Holes

- In traditional assembly, you'd see mechanical fasteners (bolts, rivets) that pierce through flanges.
- This method **weakens the structure** and **disrupts the smooth outer surface** (bad for aesthetics & aerodynamics).

Flanged Polymer Parts with Adhesive Bonding

- Instead of bolting, the yacht's panels are designed with **flanges** that fit together like puzzle pieces.
- A **solvent-based adhesive** is sprayed using a **compressed air spray gun**.
- The adhesive is applied **within the flange seam** — all around the part.
- Pressure is applied to hold parts together during curing.

Advantages:

- **No visible joins or holes** → sleek, waterproof surface.
- **Stronger joints** → avoids stress concentration from bolt holes.
- **Lightweight and corrosion-resistant** → ideal for marine applications.
- **Cleaner aesthetic** → luxury finish.

TYPES OF ADHESIVE

Types of Adhesive - Explained

1. Cyanoacrylate (Super Glue)

- **Very fast-setting**, bonds in seconds.
- Great for small parts, plastics, rubber, and skin.
- Brittle under shock or high heat.
- Cures with **moisture in the air**.

2. Epoxy

- Two-part adhesive: resin + hardener.
- **Very strong**, durable, resistant to chemicals and heat.
- Excellent for **metal, composites, and structural bonding**.
- Slow to cure (minutes to hours).

3. Polyurethane

- Flexible and durable.
- Bonds a wide range of materials: metal, wood, plastics, glass.

- Moisture-cured and often used in **construction and automotive**.

4. Silicone

- **Flexible, rubbery** adhesive.
- Excellent for sealing joints in moist or high-temperature environments.
- Common in **electronics, kitchens, and windows**.

5. Anaerobic Adhesives

- Cure in the **absence of air** (oxygen).
- Used to seal **threads, gaskets, and bearings** (e.g., Loctite).
- Great for metal-metal joins.

6. Toughened Acrylic

- Modified acrylics for improved toughness.
- Excellent **impact resistance** and strong bonding.
- Used in aerospace, automotive, and sports equipment.

7. Phenolic

- Thermosetting resin, often used as a binder in **wood products** (e.g., plywood).
- High-temperature resistance, but **brittle**.
- Used in **aerospace composites and laminates**.

8. MS Polymer (Modified Silane Polymer)

- Hybrid adhesive combining silicone and polyurethane properties.
- UV and weather resistant.
- Used in **construction and transport (e.g., trains, yachts)**.

9. Polysulphide

- Flexible, resistant to fuels and chemicals.
- Ideal for **aerospace fuel tanks**, **marine sealing**.
- Cures slowly.

10. Hot Melt

- Thermoplastic adhesive applied hot (melted), then solidifies on cooling.
- Quick bonding, used in **packaging, crafts, and furniture**.
- Low strength but fast and clean.

11. PVA (Polyvinyl Acetate)

- **White glue / wood glue**.
- Common in **woodworking, paper, and crafts**.
- Water-based and easy to use, but weak in wet or hot conditions.

12. UV Cure

- Cures (hardens) when exposed to **ultraviolet light**.
- Used in **electronics, glass, and medical devices**.
- Fast and precise with clear adhesives.

Adhesive	Key Use Area	Strength	Cure Time	Special Trait
Cyanoacrylate	Quick fixes, small parts	High	Seconds	Moisture-cured
Epoxy	Structural bonding	Very High	Slow (hrs)	Heat/chemical resistant
Polyurethane	Versatile construction	Medium	Medium	Flexible, waterproof
Silicone	Sealing in moist environments	Low	Slow (24 hrs)	Flexible, high-temp
Anaerobic	Thread locking in metals	Medium	Slow	Cures without air
Toughened Acrylic	Aerospace, impact resistance	High	Medium	Shock-resistant
Phenolic	Composites, wood panels	High	Heat required	Brittle, heatproof
MS Polymer	Building & transport	Medium	Medium	UV/weather resistant
Polysulphide	Aerospace, marine	Medium	Slow	Fuel/chemical resistant
Hot Melt	Packaging	Low	Seconds	Fast, low strength
PVA	Wood, crafts	Low	Slow	Water-soluble
UV Cure	Electronics, medical	Medium	Seconds (w/UV)	Needs UV light to cure

1. Polyurethane

- **Requirements**: Moisture to cure (some are two-part).
- **Strengths**: Flexible, impact resistant, good for bonding different materials.
- **Uses**: Automotive, construction, footwear.
- **Surface prep**: Clean, may require primer for some plastics or metals.

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2. Epoxy

- **Requirements**: Two-part mix (resin + hardener).
- **Strengths**: Very strong, chemical and heat resistant.
- **Uses**: Structural bonding in aerospace, electronics, marine.
- **Surface prep**: Clean, roughened surface improves bond.

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3. MS Polymer (Modified Silane)

- **Requirements**: Cures with moisture.
- **Strengths**: UV resistant, flexible, paintable.
- **Uses**: Construction, sealing, bonding of panels.
- **Surface prep**: Clean, dry, some primers may be used.

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4. PVA (Polyvinyl Acetate)

- **Requirements**: Water-based, cures as water evaporates.
- **Strengths**: Easy to use, non-toxic.
- **Uses**: Woodworking, paper, fabrics.
- **Surface prep**: Clean, dry, porous surfaces preferred.

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5. Polysulphide

- **Requirements**: Often two-part, cures chemically.
- **Strengths**: Flexible, fuel-resistant.
- **Uses**: Aerospace, glazing, marine sealants.
- **Surface prep**: Clean, primer may be required for some substrates.

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6. Cyanoacrylate (Super Glue)

- **Requirements**: Moisture in air to cure.
- **Strengths**: Rapid bonding, strong in tension.
- **Uses**: Plastics, rubbers, electronics, medical devices.
- **Surface prep**: Dry and clean; acidic surfaces can slow cure.

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7. UV Cure

- **Requirements**: UV light exposure to cure.
- **Strengths**: Fast cure, precision bonding.
- **Uses**: Medical, electronics, glass.
- **Surface prep**: Transparent or UV-transmitting surfaces.

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8. Anaerobic

- **Requirements**: Absence of oxygen and presence of metal ions.
- **Strengths**: Cures between metal parts (threadlockers, flange sealants).
- **Uses**: Mechanical assemblies (bolts, shafts).
- **Surface prep**: Clean metal surfaces, no large gaps.

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9. Toughened Acrylic

- **Requirements**: Two-part, or primer-activated systems.
- **Strengths**: Tough, fast, can bond oily surfaces.
- **Uses**: Automotive, aerospace, structural bonding.

- **Surface prep**: Minimal, tolerates surface contaminants.

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10. Silicone

- **Requirements**: Cures with moisture or heat.
- **Strengths**: Flexible, high-temperature and chemical resistant.
- **Uses**: Electronics, automotive, cookware, seals.
- **Surface prep**: Clean surfaces, sometimes a primer is used.

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11. Hot Melt

- **Requirements**: Heat applied to melt adhesive, cools to solidify.
- **Strengths**: Fast setting, easy application.
- **Uses**: Packaging, furniture, textiles.
- **Surface prep**: Clean and dry; mostly for porous or non-critical surfaces.

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12. Phenolic

- **Requirements**: Heat and pressure to cure (thermosetting).
- **Strengths**: Very high temperature and chemical resistance.
- **Uses**: Aerospace, electronics, industrial composites.
- **Surface prep**: Clean, often requires pressure bonding during cure

What is Surface Wetting?

Surface wetting describes **how well a liquid (like glue or paint)** spreads out on a surface. Good wetting = strong bond.

- **Higher surface energy** → **better wetting** → **stronger adhesive bonding**.
- **Low surface energy** (e.g. PTFE/Teflon) resists wetting — glue rolls

Types of Loads & Stress Distribution

Left Side: Types of Loads on Joints

These show different ways adhesive joints can be stressed:

1. **Tensile Load** (pulling apart directly) – good distribution
2. **Shear Load** (sliding past each other) – good
3. **Compressive Load** – also well distributed
4. **Peel Load** (one side lifting away) – very poor
5. **Cleavage Load** (prying open) – very poor

✅ **Good loads** (tensile, shear, compressive) spread stress evenly.
❌ **Bad loads** (peel, cleavage) create **stress peaks** at the ends, which can cause **early failure**.

Right Side: Stress Distribution Graphs

These charts show how stress is spread across the bond area:

- **Ideal**: flat line – stress evenly distributed.
- **Peel/Cleavage**: stress is **concentrated at the edge** → leads to crack initiation.

🔍 Slide 2: Design to Reduce Stress Peaks

This slide shows **bad vs. improved joint designs** when peel or cleavage loads are unavoidable.

Left Column: Peel Load Problems

- Joints stressed in peel (e.g., top flaps being peeled open).
- Leads to **concentrated stress** at the edge.

Right Column: Better Design Solutions

- **Add flanges, fillets, or overlaps** to redirect the load from peel → to shear or compression.
- **Wrap the adhesive around the load path** so it's not peeling but being sheared.
- **Use continuous bonding areas** rather than single-point joins.

✅ Key Takeaways

- **Avoid peel and cleavage** stresses in design.
- **Distribute loads across large areas**, preferably in shear or compression.
- Use smart **geometric design** (like overlaps or wraps) to **redirect forces**.
- Minimising stress peaks extends the **life and safety** of adhesive joints.

ROD AND TUBE JOINT DESIGN

Straight Butt Joint

🧩 **Use**: Good for rods in **tension and compression**.
⚠️ **Weakness**: **Poor in bending** unless the bond area is large.
✅ Simple but not good for flexing loads.

2. Angled Insertion Joint

🧩 **Use**: Good for **rods only** (not tubes).
✅ Angled insertion gives some mechanical interlock.
⚠️ Requires precise shaping.

3. Tube-in-Tube (Socket) Joint

🧩 **Use**: Good for **rods**, where one part fits inside another.
✅ Easy to align, has moderate bonding area.
⚠️ May not suit all diameters.

4. Sleeved Lap Joint

🧩 **Use**: Works well for **rods or tubes**.
✅ Balanced and stronger due to large contact area.
⚠️ Slightly more complex to fabricate.

5. Tapered Insertion Joint

🧩 **Use**: Generally **poor for tubes**.
⚠️ Tapered ends can focus stress and reduce strength.
❌ Avoid in critical structural joints.

6. Pinned Collar Joint

🧩 **Use**: Good for **rods** but **slower** to produce.
✅ Adds mechanical fixing (pin) for added strength.
⚠️ More time-consuming, better for high-strength needs.

7. Threaded End Joint

🧩 **Use**: Good for **tubes if wall thickness** allows machining.
✅ Secure and removable connection.
⚠️ Requires precision cutting and thicker walls.

8. Elbow Joint

🧩 **Use**: For **corners** in rods or tubes.
✅ Elbows offer neat 90° redirection.
⚠️ Best used with **prefabricated elbow fittings** for strength and ease.

Cylindrical Bonding

🔧 **What is it?**

- A method of **joining cylindrical parts** (e.g. gears, shafts) using **press-fit + adhesive**.
- Commonly used in **Scania truck gearboxes**, where gear sets are **press-bonded** onto a **countershaft**.

💡 **How it works:**

- One part is slightly larger and pressed into another (interference fit).
- An **anaerobic adhesive** (like Loctite) fills microscopic gaps.
- It **cures in the absence of oxygen**, forming a **solid bond**.

✅ **Advantages:**

- **Even load distribution** across the surface.
- **Prevents fretting** (microscopic movement leading to wear).
- **High torque resistance** without machining keyways or splines.

Threadsealing

🔧 **What is it?**

- Involves **applying sealant** to threaded parts (nuts, bolts, pipe threads).
- Prevents **leakage of gases or fluids** under pressure.

🛢️ **Example shown:**

- **Petrol pump thread sealing** — a **safety-critical** application.
- Uses **anaerobic adhesives** designed to:
    - Fill gaps between threads,

- Prevent leaks,
- Lock threads in place to resist vibration loosening.

✅ **Advantages:**

- **Leak-proof and vibration-proof**.
- **Chemical and pressure resistant**.
- No need for tapes or mechanical seals.

Additive Manufacturing Techniques Summary

Technique	How It Works	Key Feature	Typical Materials	Example Applications
Selective Laser Sintering (SLS)	A laser selectively sinters powdered polymer, layer by layer.	Laser sinters powder layer-by-layer (polymers)	Nylon, ABS, polycarbonate	Power tool housings, circuit breakers, aerospace handles
Selective Laser Sintering of Metals	A laser partially melts metal powder layer-by-layer. The porous part is then infiltrated with bronze.	Laser sinters metal powder, then infiltrated with bronze	Stainless steel, titanium + bronze	Injection mould tools with cooling channels, die casting moulds
Stereo-lithography (SLA)	A UV laser cures layers of liquid photopolymer resin.	Uses UV light to cure liquid photopolymer	Photopolymers (acrylates)	Phone cases, implants, EDM electrodes
Laminated Object Manufacturing (LOM)	Layers of material are bonded and laser-cut into shape.	Bond and laser-cut sheets layer by layer	Paper, polymer, metal sheets	Casting moulds, automotive mock-ups
Solid Ground Curing (SGC)	UV light cures resin through masks to form each layer simultaneously.	UV-curing resin through masks, layer by layer	Photosensitive resin, wax	Detailed prototypes, electronics enclosures
Fused Deposition Modelling (FDM)	A heated nozzle extrudes molten thermoplastic filament.	Heated nozzle extrudes thermoplastic filament	ABS, Nylon, PLA	Rapid part prototypes, functional models
Ballistic Particle Manufacturing (BPM)	Molten droplets are jetted by piezo system and solidified layer by layer.	Molten droplets jetted by piezo system and frozen	Thermoplastics	Experimental rapid prototyping
3D Printing (Inkjet-based)	Inkjet heads deposit tiny polymer	Multiple jets deposit	Thermoplastic polymer	Basic concept models and fast prints

	droplets layer by layer.	polymer like an inkjet printer		

Process	Key Method	Materials Used	How to Distinguish It	Example Use
SLS (Polymers)	Laser sintering of **polymer powders** layer by layer	ABS, Nylon, Polycarbonate	Uses powder + laser, no support material needed. Fast and accurate.	Airframe parts, housings, spotlight parts
SLS (Metals)	Laser sintering of **metal powders** + infiltrated with bronze	Stainless steel, Titanium	Same as polymer SLS but for metals. Post-infiltration with bronze.	Injection mould tools with cooling channels
SLA (Stereolithography)	UV laser cures **liquid photopolymer** layer by layer	Acrylates (photopolymers)	Liquid resin, not powder. Requires **post-curing**. Very smooth finish.	Bone implants, phone cases, EDM electrodes
LOM (Laminated Object Manufacturing)	Laser cuts and bonds **sheet layers** (paper/polymer/metal)	Paper, polymer sheets, thin metal sheets	Feeds from a roll, then laser cuts each layer. Looks like lamination.	Visual models, casting patterns
SGC (Solid Ground Curing)	UV light cures resin via a **photocopier-like mask**	Photosensitive resin	Entire layer cured at once. **Wax** supports applied and chilled.	Detailed electronic enclosures
FDM (Fused Deposition Modelling)	Heated nozzle extrudes **filament** like a glue gun	ABS, Nylon, PLA	Most common home 3D printer. Heated nozzle + filament roll.	Functional prototypes, low-cost models
BPM (Ballistic Particle Mfg.)	Piezo-jet shoots molten **microparticles** that freeze	Thermoplastics	Think of tiny droplets sprayed and frozen. White finish only.	Experimental prototyping
3D Printing (Inkjet-based)	Multiple inkjets deposit **polymer droplets**	Thermoplastics	Uses **inkjet heads** like a printer. Very fast.	Concept models, early visual prototypes